E-COMMERCE CUSTOMER BEHAVIOUR ANALYSIS

RFM Segmentation, Revenue Analysis & Business Insights

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Tools: Python | Pandas | SQL (SQLite) | Power BI

Dataset: UCI Online Retail II

1. Executive Summary

This project analyses transactional data from a UK-based online retail company to understand customer buying behaviour, identify high-value and at-risk customer segments, and uncover revenue trends across geographies and time periods.

Using Python and Pandas for data cleaning and exploratory analysis, SQL for structured querying, and Power BI for interactive dashboards, the project produces an end-to-end analysis covering 781,717 clean transactions across 5,860 unique customers and 43 countries over a 20-month period (January 2010 to December 2011).

The RFM (Recency, Frequency, Monetary) segmentation reveals that 1,801 Champions, just 31% of the customer base, generate GBP 13.2 million, representing approximately 76% of total revenue. Simultaneously, 1,107 At Risk customers with GBP 2.1 million in historical revenue have not purchased in over 200 days, representing an urgent retention opportunity.

Three actionable business recommendations are provided at the end of this report targeting customer retention, re-engagement, and revenue protection.

2. Problem Statement

An online retail company wants to understand customer buying behaviour, specifically who their most valuable customers are, who is at risk of churning, and which regions drive the most revenue.

This analysis will help the marketing and sales teams identify high-value segments to retain, re-engage at-risk customers through targeted campaigns, and allocate resources to high-performing regions, all based on RFM segmentation and revenue trends.

Business Questions This Analysis Answers

- Who are the most valuable customers and how much do they contribute to revenue?
- Which customers are at risk of churning or have already been lost?
- Which countries and regions generate the most revenue?
- What are the monthly revenue trends, are there seasonal patterns?
- Which products drive the highest volume vs highest revenue?

3. Dataset Overview

The dataset used in this project is the UCI Online Retail II dataset, sourced from Kaggle. It contains real transactional data from a UK-based online retailer selling giftware and homeware products.

Attribute	Detail	Value
Source	UCI Online Retail II (Kaggle)	-
File Format	CSV	-
Raw Rows	1,048,575	Excel row limit hit
Columns	8	Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
Date Range	Jan 2010 - Dec 2011	~20 months of transactions
Unique Countries	43	UK-based retailer, global customers

Column Descriptions

- Invoice: Unique identifier for each transaction
- StockCode: Product code for each item
- Description: Product name
- Quantity: Number of units purchased (negative = returns)
- InvoiceDate: Date and time of the transaction
- Price: Unit price in GBP
- Customer ID: Unique customer identifier (blank for guest checkouts)
- Country: Country of the customer

4. Data Cleaning

Data cleaning was performed using Python and the Pandas library. The raw dataset contained several quality issues that required resolution before any analysis could be conducted. Each decision was made based on the business context of the analysis.

Cleaning Step	Rows Before	Rows After	Removed	Reason
Raw Dataset	1,048,575	-	-	Starting point
Drop missing Customer ID	1,048,575	811,893	236,682	No ID = unusable for RFM
Remove negative Quantity	811,893	793,380	18,513	Returns/cancellations
Remove negative Price	793,380	793,309	71	Corrupt data
Remove duplicates	793,309	781,717	11,592	Duplicate transactions
Final Clean Dataset		**781,717**		**9 columns**

Key Cleaning Decisions

- Missing Customer ID (236,682 rows removed): The entire RFM analysis depends on customer identity. Transactions with no Customer ID cannot be attributed to any customer and are therefore useless for segmentation. This is a deliberate and documented business decision.
- Negative Quantities (18,513 rows removed): Negative quantities represent product returns and cancellations. Including these would distort purchase frequency and revenue calculations.
- Negative Prices (71 rows removed): These represent data entry errors or system-generated adjustments and have no valid business interpretation.
- Duplicate Rows (11,592 rows removed): Exact duplicate records would cause double-counting in all aggregations.
- TotalPrice Column Added: A new column was created as Quantity x Price to represent the actual revenue contribution of each transaction row, the core metric for Monetary value in RFM.

5. Exploratory Data Analysis (EDA)

Following data cleaning, exploratory analysis was conducted to understand patterns in revenue, geography, customer behaviour, and product performance. Four charts were generated using Matplotlib and Seaborn.

Finding 1: United Kingdom Dominates Revenue

The UK accounts for the overwhelming majority of total revenue at GBP 14.3 million, approximately 82% of the total. EIRE (Ireland) is the second-largest market at GBP 615K, followed by Netherlands at GBP 542K. Non-UK markets collectively represent a significant growth opportunity, particularly Germany (GBP 423K, 106 customers) and France (GBP 350K, 94 customers) which show strong revenue despite small customer counts.

Finding 2: Clear Seasonal Revenue Pattern

Monthly revenue shows a recurring seasonal pattern across both 2010 and 2011. Revenue rises steadily from mid-year, peaks sharply in October-November (consistent with pre-Christmas retail demand), then drops significantly in December-January. The November 2011 peak reached GBP 1.16 million, the highest single month in the dataset. The sharp December 2011 drop to GBP 99K is likely due to incomplete data for that month rather than actual sales decline.

Finding 3: Single Customer Revenue Concentration Risk

Customer 18102 alone generated GBP 597,336 from 142 orders, more than any other customer. The top 10 customers collectively account for GBP 2.75 million. This level of concentration in a small number of high-value customers represents a business risk: losing even one or two of these customers would have a measurable revenue impact.

Finding 4: Volume Leaders vs Revenue Leaders Differ

The top product by quantity sold (World War 2 Gliders - 107,860 units) generated only GBP 24,552 in revenue. In contrast, White Hanging Heart T-Light Holder generated GBP 250,172 despite lower volume. This gap between volume and revenue performance is a key insight for inventory and pricing strategy.

6. RFM Analysis

What is RFM?

RFM is a customer segmentation methodology based on three behavioural metrics:

- Recency (R): How many days since the customer last made a purchase. Lower recency = more engaged customer.
- Frequency (F): How many unique invoices/orders the customer has placed in total. Higher frequency = more loyal customer.
- Monetary (M): Total amount spent by the customer across all transactions. Higher monetary value = higher business value.

Each customer was scored 1-4 on each dimension using quartile-based scoring, then assigned to one of seven business segments based on their combined RFM profile.

RFM Segment Results

Segment	Customers	Avg Recency	Avg Frequency	Avg Spend (GBP)	Total Revenue
Champions	1,801	30 days	14 orders	7,347	13,232,279
At Risk	1,107	223 days	5 orders	1,926	2,131,664
Lost	1,428	458 days	1 order	290	414,756
Loyal Customers	788	37 days	3 orders	765	603,127
Hibernating	229	461 days	5 orders	1,983	454,166
Cannot Lose Them	163	450 days	2 orders	1,860	303,204
Potential Loyalists	344	45 days	1 order	375	128,893

Critical Business Insights from RFM

Insight 1 - Champions drive the entire business:

1,801 Champions (31% of customers) generate GBP 13.2 million, approximately 76% of total revenue. With an average recency of just 30 days and 14 orders per customer, these are highly active, high-value buyers. The business is heavily dependent on this segment.

Insight 2 - At Risk segment is a revenue emergency:

1,107 customers with GBP 2.1 million in cumulative spend have gone 223 days without purchasing. These customers were previously buying regularly (average 5 orders). This is not natural churn; it is a retention failure that can be addressed with targeted re-engagement campaigns.

Insight 3 - Cannot Lose Them requires immediate action:

Only 163 customers but with the second-highest average spend at GBP 1,860 per customer. They have been inactive for 450 days on average. These are the highest-priority recovery targets, high value, clearly disengaged, still reachable.

7. SQL Analysis

Five SQL queries were written using SQLite (implemented via Python's sqlite3 library) to extract structured business insights from the cleaned dataset. Both the transactions and RFM tables were loaded into the database.

#	Query	Business Question Answered
1	Revenue by Country	Which countries generate the most revenue and have the most unique customers?
2	Monthly Revenue Trend	How does revenue change month over month? Are there seasonal patterns?
3	Segment Revenue Summary	How much revenue and how many customers fall in each RFM segment?
4	Top 10 Customers by Revenue	Who are the highest value individual customers by total spend?
5	Top 10 Products by Quantity	Which products sell the highest volume vs generate the most revenue?

Key SQL Findings

- Query 1 confirmed UK dominance with GBP 14.3M from 5,337 unique customers, more than 10x the next market.
- Query 2 revealed the seasonal November peak pattern recurring in both 2010 (GBP 1.17M) and 2011 (GBP 1.16M).
- Query 3 validated the Python RFM output, Champions segment confirmed at GBP 13.2M total revenue.
- Query 4 highlighted a critical insight: Customer 14911 placed 391 orders but generated only GBP 289K, while Customer 18102 placed 142 orders and generated GBP 597K. Order count does not equal revenue, order value matters more.
- Query 5 showed that the top quantity product (World War 2 Gliders) and top revenue product (White Hanging Heart T-Light Holder) are completely different, volume and value leadership do not overlap.

8. Power BI Dashboard Summary

An interactive three-page Power BI dashboard was built using the cleaned data and SQL query outputs. All pages share a consistent visual theme.

Page 1: Retail Sales Overview

- 4 KPI Cards: Total Revenue (GBP 17.27M), Total Orders (36,457), Total Customers (5,860), Countries Served (41)
- Monthly Revenue Trend Line Chart: Jan 2010 to Dec 2011, showing seasonal peaks
- Top 10 Countries by Revenue Bar Chart: UK dominance clearly visible
- Revenue by Geography Map: Bubble map showing customer concentration in Europe

Page 2: Customer Segment Analysis

- Customer Count by Segment Bar Chart: Champions largest segment, Lost second largest
- Revenue by Segment Bar Chart: Champions revenue bar 5x longer than all others combined
- Customer Segment Distribution Donut Chart: Champions = 30.73% of customer base
- Segment Summary Table: All 7 segments with Customer Count, Avg. Monetary, Total Revenue

Page 3: Product and Customer Analysis

- Top 10 Products by Quantity Sold: Volume leaders identified
- Top 10 Products by Revenue: Revenue leaders identified, different from volume leaders
- Top 10 Customers by Revenue Bar Chart: Customer 18102 clearly the highest value
- Top Customer Details Table: Customer ID, Total Orders, Total Revenue for top 10

9. Business Recommendations

Based on the RFM segmentation, revenue analysis, and customer behaviour patterns identified in this project, three actionable business recommendations are proposed:

Recommendation	Action	Expected Impact
1. Protect Champions	Launch a loyalty programme with exclusive offers, early access, and personalised communication targeting the 1,801 Champions segment.	Retaining even 90% of Champions protects GBP 11.9M in annual revenue.
2. Re-engage At Risk	Deploy a targeted email re-engagement campaign with personalised discounts for the 1,107 At Risk customers who have not purchased in 223 days on average.	Recovering 20% of this segment = GBP 426K in recovered revenue.
3. Recover Cannot Lose Them	Initiate immediate personal outreach, phone calls, exclusive win-back offers - for the 163 Cannot Lose Them customers who averaged GBP 1,860 spend but have been inactive for 450 days.	High-value recovery priority. Even 30% recovery = GBP 90K from a tiny segment.

Additional Recommendations

- Seasonal Inventory Planning: November is consistently the peak revenue month. The business should increase inventory levels and marketing spend from September to capture the full pre-Christmas demand window.
- International Expansion: Germany (106 customers, GBP 423K) and France (94 customers, GBP 350K) show strong revenue per customer ratios. A targeted international marketing campaign in these markets could yield significant revenue growth with relatively small customer acquisition numbers.
- Volume vs Value Product Strategy: High-volume, low-revenue products like World War 2 Gliders should be reviewed for pricing optimisation. High-revenue products like White Hanging Heart T-Light Holder should be prioritised in inventory and promoted more aggressively.

10. Conclusion

This project demonstrates a complete end-to-end data analysis workflow, from raw data acquisition and cleaning through exploratory analysis, RFM segmentation, SQL querying, and interactive dashboard development, applied to a real-world e-commerce dataset.

The central finding is that this retail business has a highly concentrated revenue base: a small group of Champions customers generates the vast majority of revenue, while a significant segment of previously active customers is drifting toward churn without intervention. The data provides clear, actionable signals for marketing and retention strategy.

The combination of Python, SQL, and Power BI used in this project reflects a standard Data Analyst toolkit and demonstrates the ability to translate raw transactional data into business decisions.

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